

Strategic Recovery & Capital Optimizer (PoC)

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Target Stakeholders: Chief Risk Officers (CRO), CFOs, & Basel IV Steering Committees

Executive Summary

The **Strategic Recovery & Capital Optimizer** is an executive-grade decision-support portal designed to navigate the complexities of the modern regulatory landscape. By unifying recovery operations with capital adequacy metrics (CET1), this platform demonstrates how high-performance recovery strategies directly mitigate **Capital Drag** and drive shareholder value.

Strategic Business Value:

- **CET1 Ratio Optimization:** Simulates the reduction of Risk-Weighted Assets (RWA) through increased recovery yields, effectively "freeing up" Tier 1 capital for reinvestment.
- **ROTCE Pathfinding:** Provides a direct line of sight toward **Return on Tangible Common Equity (ROTCE)** targets (e.g., 11% by 2026), proving that recovery efficiency is a primary driver of bottom-line profitability.
- **Macro-Resilience:** Stress-tests Net Charge-Off (NCO) projections across a 9-quarter horizon against "Severely Adverse" economic scenarios.

Strategic Modules & Executive UX

The portal features a "Human-in-the-Loop" architecture, protected by an **Executive Gateway** for secure leadership review.

1. Capital Report & WARR Analysis

- **Goal:** Track the yield health of Branded Cards and Retail segments.

- **Analytics:** Dynamic modeling of **Weighted Average Recovery Rates (WARR)** with automated Year-over-Year (YoY) variance tracking to identify performance leakage and its impact on CET1 ratios.

2. Dynamic Stress Testing (9-Quarter Engine)

- **Goal:** Benchmark institutional resilience against peer institutions (e.g., JPM, Amex).
- **Simulation:** An NCO projection engine that overlays macro-economic severity sliders (Unemployment, GDP), allowing leaders to visualize capital requirements in recessionary environments.

3. Vendor Governance & Placement Simulator

- **Sunburst Analytics:** Visualizes the relationship between Tiered vendors and **Capital Drag (bps)**.
- **Inventory Swap Simulator:** A "what-if" tool to calculate the net recovery gain of shifting inventory between agencies or moving to legal-track recovery.
- **Jurisdictional Heatmap:** A Choropleth-based tracker monitoring state-level regulatory exceptions and audit updates (e.g., CCPA audits in CA).



Technical Architecture

This project demonstrates proficiency in financial engineering, high-fidelity data visualization, and secure web deployment.

Layer	Technical Implementation
UI/UX Framework	Streamlit with custom CSS-in-Python styling for an "Executive Gateway" interface.
Financial Engine	High-performance <code>@st.cache_data</code> logic for simulating CET1 basis point (bps) impacts across multi-million row dataframes.
Visualizations	Plotly Graph Objects for financial curves and Plotly Express for sunburst/geospatial analytics.

Strategic KPI Glossary

Key metrics calculated within the PoC are aligned with standard banking and recovery operations:

- **CET1 Ratio:** The primary measure of a bank's financial strength; recovery lifts reduce RWA to boost this ratio.
- **ROTCE:** Return on Tangible Common Equity; the ultimate yardstick for shareholder value and profitability.
- **Capital Drag (bps):** The "cost of carry" for inventory measured in basis points, reflecting capital tied up during long liquidation timelines.
- **Yield Premium:** The additional recovery percentage gained by utilizing higher-cost legal channels over standard collections.

Installation & Usage

To launch the decision engine locally:

1. **Clone the repository:**

```
git clone https://github.com/your-username/strategic-recovery-optimizer.git
```

2. **Install dependencies:**

```
pip install -r requirements.txt
```

3. **Run the application:**

```
streamlit run app.py
```

Executive Gateway Access Code: Recovery2026

Repository Strategy

This repository serves as a **Narrative of Authority**, shifting the focus from simple operations to institutional capital management.

Operational Focus (Old)	Capital Strategy Focus (New)
Lists "Vendor Placement" as a feature.	Frames it as "Managing Capital Drag (bps)" for the CFO.

Screenshots

Executive Hub

Logout

Executive Capital Report

Peer Strategy & Stress

Vendor Management

Regulatory Heatmap

LEAVE-BEHIND

Executive Capital & Shareholder Value Report

Target Yield Optimization Lift (%)

CET1 Capital Ratio

13.2%

↑ +1.25 bps

2026 ROTCE Target

10.5%

↑ On Track

RWA Optimization

\$165M

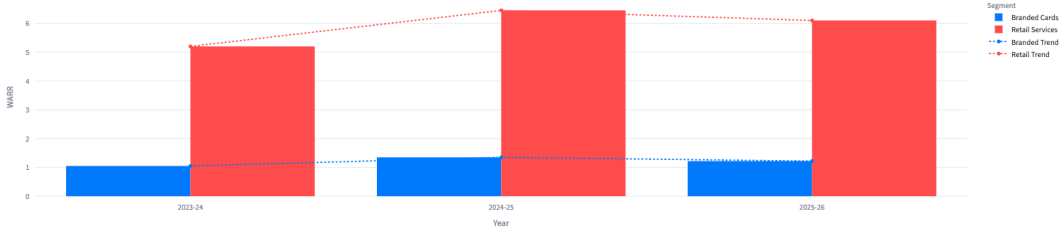
↑ Net Recovery Impact

Economic Profit

\$1.2B

↑ +5.2% YoY

Historical WARR Risk Migration with Risk Trendlines



Executive Hub

Logout

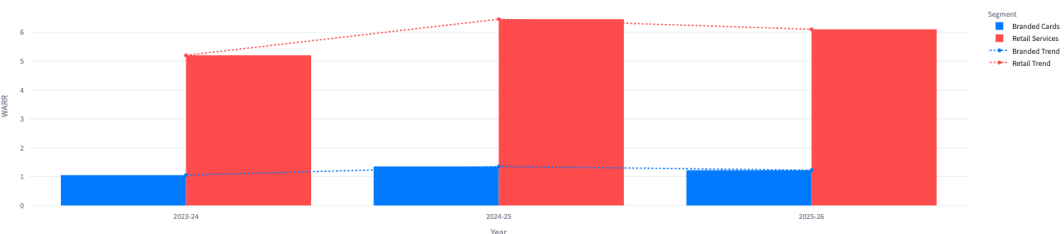
↑ +1.25 bps

↑ On Track

↑ Net Recovery Impact

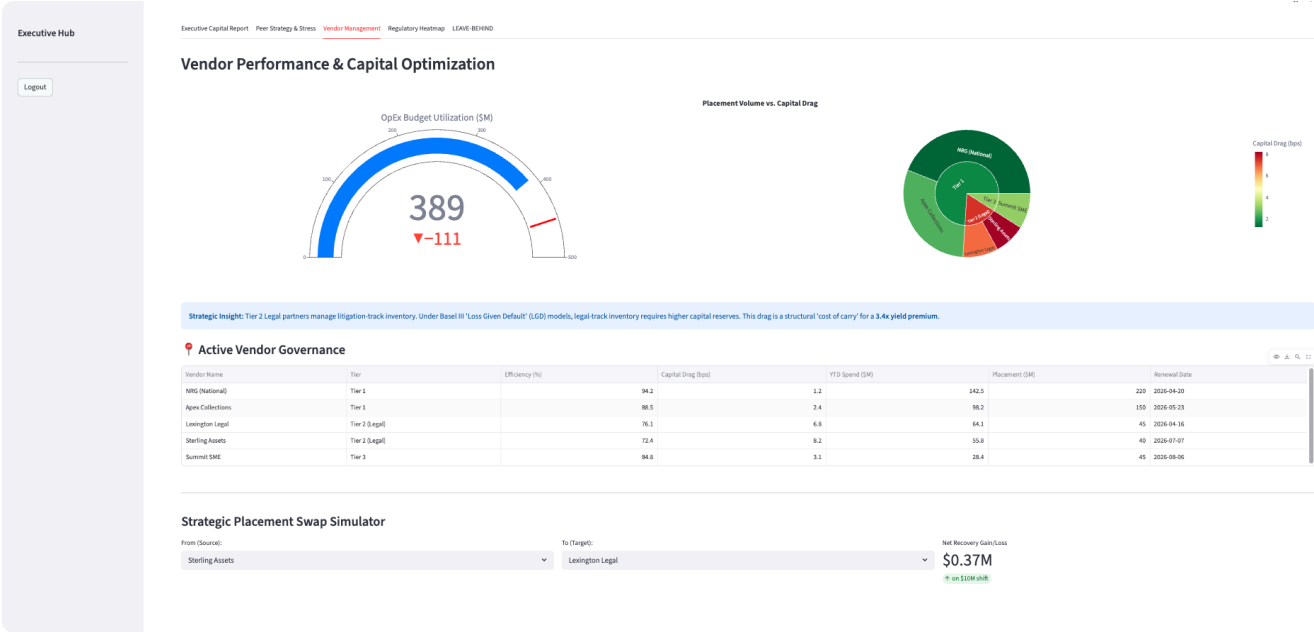
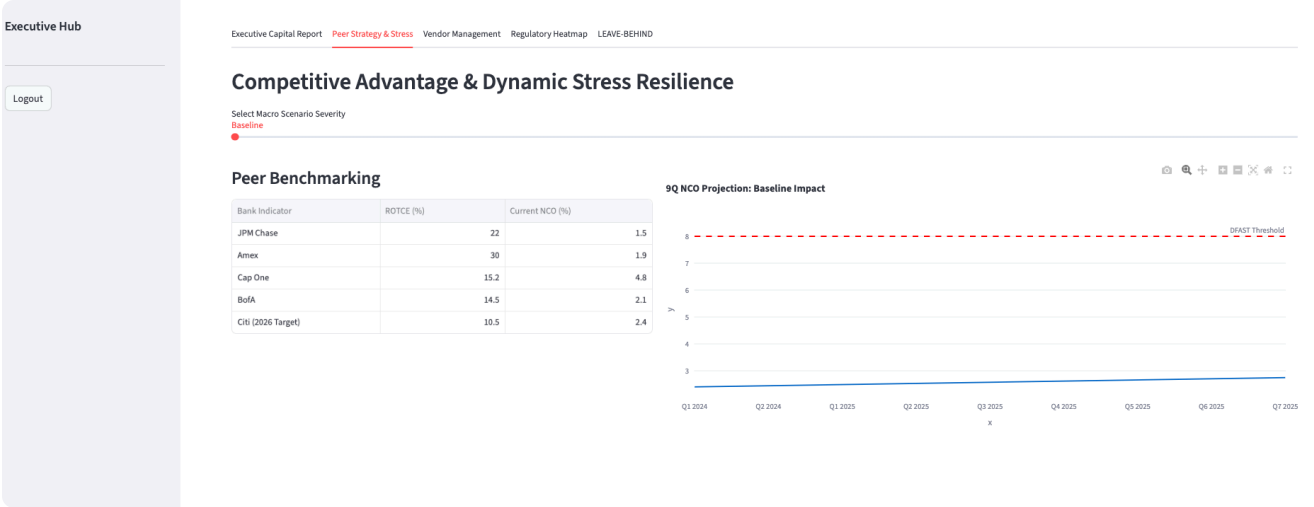
↑ +5.2% YoY

Historical WARR Risk Migration with Risk Trendlines



Historical Risk Data & YoY Variance

Year	Branded Cards	Retail Services	Branded YoY %	Retail YoY %
2023-24	1.050000	5.200000	+nan%	+nan%
2024-25	1.350000	6.450000	+28.57%	+24.04%
2025-26	1.220000	6.100000	-9.63%	-5.43%





Strategic Category	Detail
Jurisdiction	Federal
Risk Level	🟡 Moderate
Focus Area	OCC Oversight
Total Exception Volume	52 Active Exceptions
Status	Resource Review terminated Dec 2025.

Strategic Positioning Summary (Q1 2026)

- **Capital Impact:** Yield optimization adds 1.25 bps to CET1 headroom.
- **Historical Context:** Proactive 2024 optimization stabilized risk velocity in Branded Cards.

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